

Cognitive Routing of AI Agent Requests via Contextual Knowledge Building

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Motivation

- Evolving AI agents within a shared/distributed environment
 - Need to interact with each other grows with the scale and complexity of the task being accomplished.
- Single agents evolve through updating their pre-defined skills [[WikiSkill](#)], but in multi-agent systems learning is not only from external knowledge sources (like repository, files, etc.), but also through intra-agent interaction and flow of information.
- Another motivational parallel could be drawn from epistemic exchange of information among humans — a patient consulting a doctor for a medical diagnosis.
- Current approaches involve a single central orchestrating layer at the top to direct the flow of communication and work distribution among agents — [AgentOrchestra](#), [CrewAI](#)

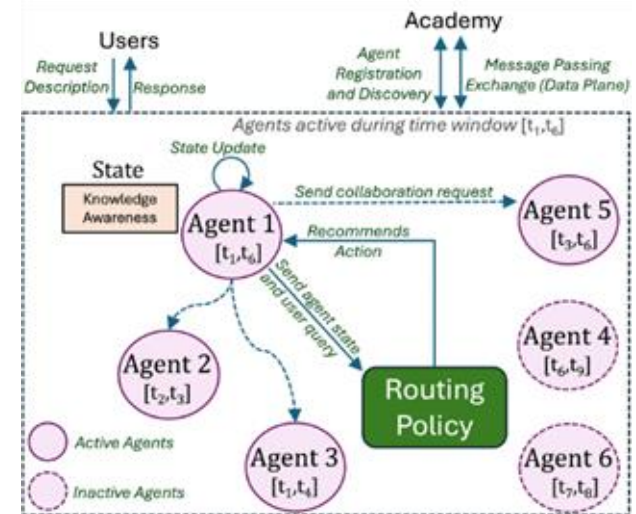


[Source](#)

How can multiple agents come together in an efficient way to solve questions which they could never do as individual units?

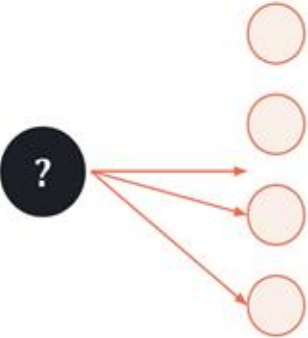
Proposal

- A decentralized group of agents that can help answer a disparate set of users' queries when no single agent can answer all the queries.
- Agents decide whether to answer the query themselves or forward it to a specific agent or group of agents that is discoverable.
- But how do these agents decide when to forward or broadcast requests?
 - Random forwarding can result in delays in getting the right answer
 - Indiscriminate broadcast increases bandwidth and LLM costs exponentially
 - No forwarding of requests results in poor query response.
- Need for a Common Decentralized Routing Policy -> Uniform across the Agent



Core Challenge

Not Knowing Who to Ask Means Asking Everyone



one question $\rightarrow$ N calls

Cost

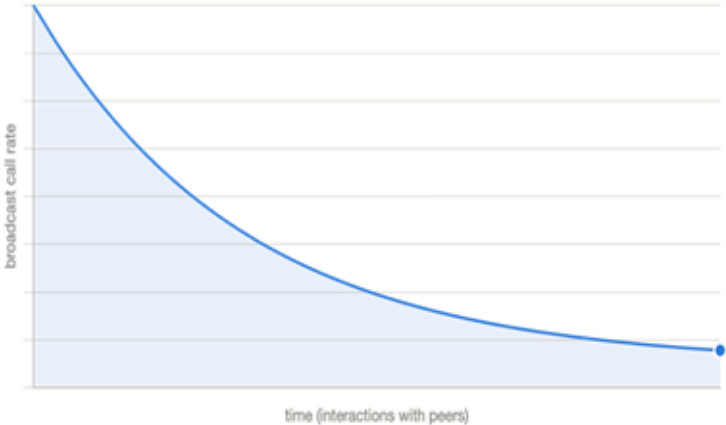
Every peer runs inference on a question only one of them could answer.

Latency

The user waits on the slowest responder, plus an aggregation pass on top.

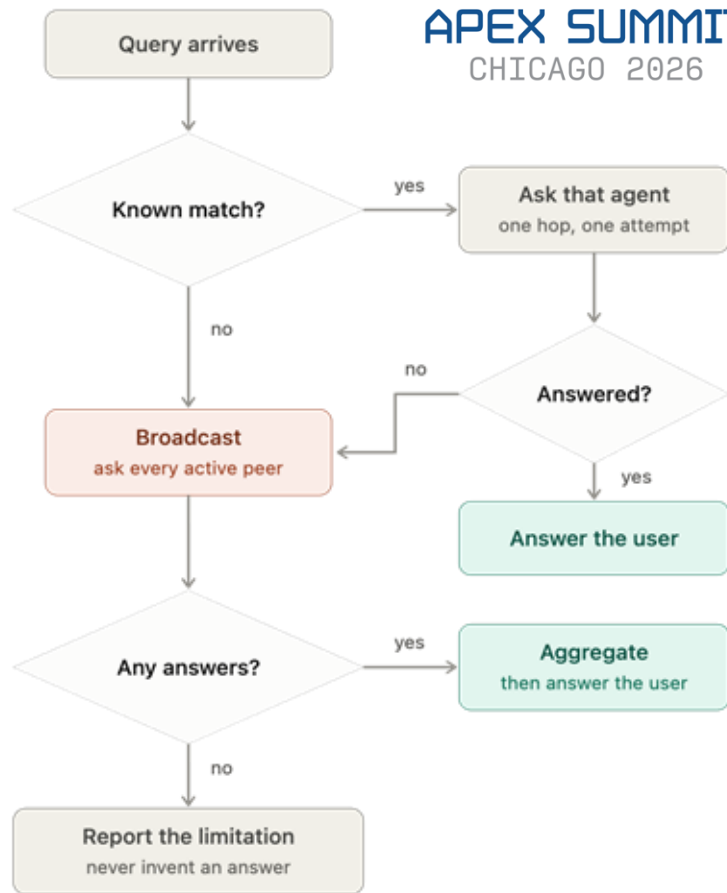
Answer quality

More speculative responses in the mix — more surface for a wrong or empty answer to survive aggregation.



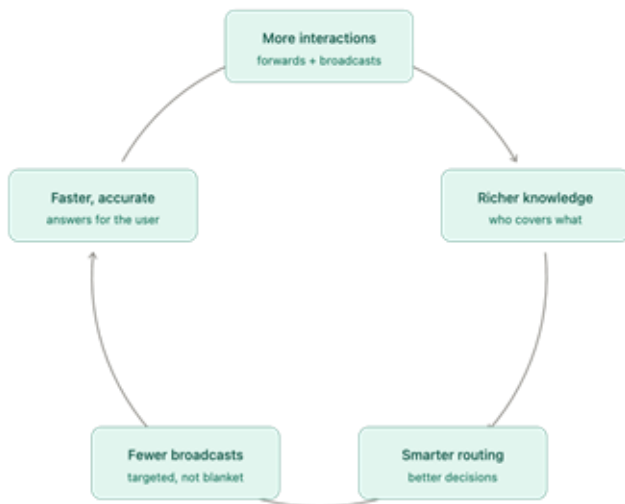
Routing Decision Flow

- When a new query arrives, a semantic decision is made whether specific agent(s) [itself included] exist that are better suited to handling that response.
- If yes, answer/forward the query.
- If no, or if the return responses are negative, make a broadcast to every available agent discoverable in range.
- If more than one response is received, we aggregate the response with right attribution.
- Each response is also used to update the knowledge-base of its peers for the original forwarding agent.
- Report the limitation if everyone fails to answer.



Knowledge Building Cycle

- As the number of interactions [query forwards/broadcasts] increase over time
- It leads to richer knowledge of peer agents capabilities
- Which results in more intelligent routing and decision-making
- Creating fewer broadcast calls
- Leading to faster TAT and accurate responses for the user.



Thank You!

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